

DAREN SHAMOUN

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SUMMARY

Software engineer building production systems end-to-end across full-stack web, mobile, and embedded targets who treats agentic AI workflows as core engineering infrastructure, utilizing persistent memory architectures, parallel agent orchestration, and prompting practices that hold production-grade quality. Recent work at BD shipped BD-Pixie, a wireless platform for the BD Alaris infusion pump deployed onto the device's embedded Linux board with paired mobile app. Graduating May 2026 Magna Cum Laude from the University of San Diego, Shiley-Marcos School of Engineering, with departmental honors and a focus in AI and Data Science (GPA 3.83).

EXPERIENCE

Software Engineer Intern — BD (Becton Dickinson)

Aug 2025 – May 2026

San Diego, CA · BD-Pixie Capstone, USD Senior Design × BD

- Led development of BD-Pixie, BD's universal mobile companion. This wireless platform enables biomedical technicians to triage and provision BD Alaris infusion pumps in place at the bedside or in the shop, eliminating the cart-laptop-cable workflow currently required to service them. Built around the biomed use case with active research into nurse and pharmacy workflows. Authored 80%+ of project commits across both ends of the system.
- Built the React Native mobile application end-to-end, including biomed UX, secure QR-code pairing, real-time pump status, on-the-spot diagnostic log retrieval over chunked BLE with end-to-end CRC integrity verification (~25–50× faster than conventional serial reads), and parallel batch provisioning of 10+ pumps (identity, network config, certificates) from a single handheld session.
- Designed and built the Python BLE server end-to-end on the pump's embedded Linux board: custom GATT communication protocol, device-side state management, and integration with the PCU's internal APIs. Refactored along SOLID principles for clean handoff to BD's firmware team for production deployment.
- Force-multiplied output through a deliberate agentic AI methodology that combines specialized parallel agents (Planner, Reviewer, Build-Error-Resolver, Security) with a project CLAUDE.md defining sacred files, layer boundaries, and BLE service contracts. Mandatory audit gates, root-cause-only fixes, and human review on commits to main. Methodology codified in an internal engineering write-up.
- Projected fleet impact (BD estimates): ~90 minutes saved per pump per service cycle, ~750 hours back per 500-pump fleet per cycle, and 50% reduction in pumps pulled from service for diagnostics.

Full-Stack Developer — Red Carpet Real Estate

Sep 2022 – Present

San Diego, CA

- Led the end-to-end digital transformation of a multi-property real estate business; solo-built a comprehensive web application that replaced paper-based operations for commercial and residential property management.
- Built the platform on Next.js + React, Flask (Python), Tailwind CSS, and PostgreSQL; architected modular data models for properties, units, tenants, leases, and financial transactions with secure document handling.

Co-Founder & Head of IT — Dashbyte

Jul 2020 – Jan 2024

San Diego, CA

- Co-founded an IT services company as a teenager; delivered 10+ custom websites for small-business clients, built and deployed ~50 computer systems (PCs, servers, cryptocurrency mining rigs), and provided ongoing IT support and network services.

EDUCATION

Bachelor of Science, Computer Science — University of San Diego

Jan 2025 – May 2026

Shiley-Marcos School of Engineering | GPA: 3.83 | Focus: AI and Data Science

- Magna Cum Laude with Departmental Honors in Computer Science.
- Eta Kappa Nu (HKN), IEEE Honor Society — Lifetime Member.
- Association for Computing Machinery (ACM) — Member.
- Senior coursework: Algorithms, Artificial Intelligence, Data Science, Object-Oriented Software Design, Networking, Digital Hardware, Automata.

Associate of Science, Computer Science — San Diego City College

Jul 2023

Full-Stack Software Developer Professional Certification — IBM

Sep 2024

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, HTML, CSS, Bash

Frameworks: React, React Native, Next.js, Node.js, Express.js, Flask, Django, Three.js, Tailwind CSS

Infrastructure: Git, Docker, Linux, PostgreSQL, REST APIs, GraphQL, Embedded Linux, Bluetooth Low Energy

AI Development: Claude Code, Agentic AI Workflows, Anthropic Claude

Engineering: System Architecture, Software Design, OOP, SOLID, Test-Driven Development, Agile / Scrum

ACTIVITIES

NCAS Aerospace Scholar — NASA

Jul 2023 – Jan 2024

- Researched NASA technologies and missions; worked as a Computer Engineer on a simulated Mars mission, contributing to a research paper and a team robotics competition.